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INTERCONNECTIONS TO PACKAGE SUBSTRATES FOR INTERCONNECT BRIDGES

Abstract

Semiconductor package substrates and assemblies comprising semiconductor package substrates are provided. The semiconductor package substrates can include interconnect bridges having through-bridge vias. The semiconductor package substrates can also include collapsible cavities in dielectric layers. The interconnect bridges can include protruding conductive pads that form electrical connections with the semiconductor package substrate. Assemblies include semiconductor package substrates and semiconductor devices.

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Background/Summary

FIELD

[0001] Descriptions are generally related to semiconductor manufacture, and more particular descriptions are related to package substrates for semiconductor devices and package substrates that include interconnect bridges.

BACKGROUND

[0002] Semiconductor chips are central to intelligent devices and systems, such as personal computers, laptops, tablets, phones, servers, and other consumer and industrial products and systems. Manufacturing semiconductor chips presents a number of challenges and these challenges are amplified as devices become smaller and performance demands increase. Challenges include, for example, unwanted material interactions, precision and scaling requirements, power delivery requirements, limited failure tolerance, and material and manufacturing costs.

[0003] Form factor miniaturization and increased levels of integration for high performance semiconductor devices are driving the need for sophisticated packaging approaches in the semiconductor industry. Die partitioning enables small form factors and high performance. Die partitioning can also improve semiconductor chip manufacturing yields but it can require fine pitched die-to-die interconnections. Interconnect bridges that can be embedded in package substrates can enable a lower cost and simpler 2.5D packaging approach (that employs an interconnect bridge in the package substrate) for very high-density interconnects between heterogeneous dies on a single package. An interconnect bridge can be embedded in a semiconductor device package substrate, and can enable very high density die-to-die connections only where needed. Interconnect bridges can be, for example, Embedded Multi-die Interconnect Bridges (EMIBs). Standard flip chip assembly can be used for robust power delivery interconnections and to connect high-speed signals directly from chip to the package substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The figures are provided to aid in understanding the invention. The figures can include diagrams and illustrations of exemplary structures, assemblies, data, methods, and systems. For ease of explanation and understanding, these structures, assemblies, data, methods, and systems, the figures are not an exhaustively detailed description. The figures therefore should not be understood to depict the entire metes and bounds of structures, assemblies, data, methods, and systems possible without departing from the scope of the invention. Additionally, features are not necessarily illustrated relatively to scale due in part to the small sizes of some features and the desire for clarity of explanation in the figures.

[0005] FIGS. 1A-1C show a semiconductor package substrate and assemblies that include the semiconductor package substrate.

[0006] FIGS. 2A-2B illustrate regions of a semiconductor package substrate.

[0007] FIGS. 3A-3C provide an additional semiconductor package substrate and assemblies that include the semiconductor package substrate.

[0008] FIGS. 4A-4C show an additional semiconductor package substrate and assemblies that include the semiconductor package substrate.

[0009] FIG. 5 illustrates an exemplary multi-chip package in which the package incorporates

interconnect bridges.

[0010] FIGS. 6A-6F illustrate a method for manufacturing a semiconductor package substrate comprising an interconnect bridge.

[0011] FIGS. 7A-7G illustrate additional methods for manufacturing a semiconductor package substrate comprising an interconnect bridge.

[0012] FIG. 8 provides an additional method for manufacturing a semiconductor package substrate comprising an interconnect bridge.

[0013] FIG. 9 provides an exemplary computing system.

[0014] Descriptions of certain details and implementations follow, including non-limiting descriptions of the figures, which depict some examples and implementations.

DETAILED DESCRIPTION

[0015] References to one or more examples are to be understood as describing a particular feature, structure, or characteristic included in at least one implementation of the invention. The phrases “one example” or “an example” are not necessarily all referring to the same example or embodiment. Any aspect described herein can potentially be combined with any other aspect or similar aspect described herein, regardless of whether the aspects are described with respect to the same figure or element.

[0016] The words “connected” and/or “coupled” can indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other and are instead separated by one or more elements but they may still co-operate or interact with each other, for example, physically, magnetically, or electrically.

[0017] The words “first,” “second,” and the like, do not indicate order, quantity, or importance, but rather are used to distinguish one element from another. The words “a” and “an” herein do not indicate a limitation of quantity, but rather denote the presence of at least one of the referenced items. The terms “follow” or “after” can indicate immediately following or following some other event or events. Other sequences of operations can also be performed according to alternative embodiments. Furthermore, additional operations may be added or removed depending on the application.

[0018] Disjunctive language such as the phrase “at least one of X, Y, or Z,” is used in general to indicate that an element or feature, may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, this disjunctive language should be understood not to imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

[0019] Flow diagrams as illustrated herein provide examples of sequences of various process actions. The flow diagrams can indicate operations to be executed by a software or firmware routine, as well as physical operations. Physical operations can be performed by semiconductor processing equipment. Although shown in a particular sequence or order, unless otherwise specified, the order of the actions can be modified. Thus, the illustrated diagrams should be understood only as examples, and the process can be performed in a different order, and some actions can be performed in parallel. Additionally, one or more actions can be omitted and not all implementations will perform all actions.

[0020] Various components described can be a means for performing the operations or functions described. Each component described can include software, hardware, or a combination of these. Some components can be implemented as software modules, hardware modules, special-purpose hardware (for example, application specific hardware, application specific integrated circuits (ASICs), digital signal processors (DSPs), etc.), embedded controllers, or hardwired circuitry). Other components can be semiconductor processing equipment that is able to perform physical operations such as, for example, lithography, chemical or laser etching, material deposition (for example, chemical vapor deposition, atomic layer deposition, electrochemical deposition, and/or sputtering), chemical mechanical polishing, and/or pick-and-place operations.

[0021] To the extent various computer operations or functions are described herein, they can be described or defined as software code, instructions, configuration, and/or data. The software content can be provided via an article of manufacture with the content stored thereon, or via a method of operating a communication interface to send data via the communication interface. A machine-readable storage medium can cause a machine to perform the functions or operations described. A machine-readable storage medium includes any mechanism that stores information in a tangible form accessible by a machine (e.g., computing device), such as recordable/non-recordable media (e.g., read only memory (ROM), random access memory (RAM), magnetic disk storage media, optical storage media, flash memory devices). Instructions can be stored on the machine-readable storage medium in a non-transitory form. A communication interface includes any mechanism that interfaces to, for example, a hardwired, wireless, or optical medium to communicate to another device, such as, for example, a memory bus interface, a processor bus interface, an Internet connection, a disk controller.

[0022] Terms such as chip, die, IC (integrated circuit) chip, IC die, microelectronic chip, microelectronic die, semiconductor die, semiconductor device, and/or semiconductor chip are interchangeable and refer to a semiconductor device comprising integrated circuits.

[0023] Semiconductor chip manufacturing processes are sometimes divided into front end of the line (FEOL) processes and back end of the line (BEOL) processes. Electronic circuits and active and passive devices within the chip, such as for example, transistors, capacitors, resistors, and/or memory cells, are manufactured in what can be referred to as FEOL processes. Memory cells include, for example, electronic circuits for random access memory (RAM), such as static RAM (sRAM), dynamic RAM (DRAM), read only memory (ROM), non-volatile memory, and/or flash memory. FEOL processes can be, for example, complementary metal-oxide semiconductor (CMOS) processes. BEOL processes include metallization of the chip where interconnects are formed in layers and the feature size of the interconnect increases in layers nearer the surface of the semiconductor chip. Interconnects in, for example, semiconductor chips that are integrated into heterogeneous packages (such as, for example, packages that include memory and logic chips), can also include through silicon vias (TSVs) that transverse the semiconductor chip device region. Semiconductor devices that have TSVs can blur distinctions between BEOL and FEOL processes.

[0024] Semiconductor chip interconnects can be created by forming a trench or through-layer via by etching a trench or via structure into a dielectric layer and filling the trench or via with metal. Dielectric layers can comprise, for example, low-K dielectrics, SiO₂, silicon nitride (SiN), silicon carbide (SiC), and/or silicon carbonitride (SiCN). Low-K dielectrics include for example, fluorine-doped SiO₂, carbon-doped SiO₂, porous SiO₂, porous carbon-doped SiO₂, combinations for the foregoing, and also these materials with gas-filled gaps. Dielectric layers that include conducting features can be intermetal dielectric (ILD) features.

[0025] The terms “package,” “packaging,” “IC package,” or “chip package,” “microelectronics package,” or “semiconductor chip package” are interchangeable and generally refer to an enclosed carrier of one or more dies, in which the dies are attached to a package substrate and encapsulated. The package substrate provides electrical interconnects between the die(s) and other dies and/or a motherboard or other printed circuit board for I/O (input/output) communication and power delivery. A package with multiple dies can, for example, be a system in a package.

[0026] A package substrate generally includes dielectric layers or structures having conductive structures on, through, and/or embedded in the dielectric layers. The dielectric layers can be, for example, build-up layers. Dielectric build-up materials include, polyimides and Ajinomoto build-up films (ABFs), although other dielectric materials are possible. Semiconductor package substrates can have cores or be coreless. Semiconductor packages having cores can have dielectric layers such as build-up layers on more than one side of a core, such as on two opposite sides of a core. Cores can include through-core vias that contain a conductive material. Other structures or devices are also possible within a package substrate.

[0027] A “core” or “package core” generally refers to a layer usually embedded within a package substrate. The core can provide structure or stiffness to a package substrate. A core is an optional feature of a package substrate. The core can be a dielectric organic or inorganic material and may have conductive vias extending through the layer. The conductive vias can include a metal, for example, copper. A package core can, for example, be comprised of a glass material (such as, for example, aluminosilicate, borosilicate, alumino-borosilicate, silica, and fused silica), silicon, silicon nitride, silicon carbide, gallium nitride, or aluminum oxide. In some examples, core materials are glass-fiber reinforced organic resins such as epoxy-based resins. A further example package substrate core is FR4 (woven glass fiber reinforces epoxy). In other examples, package substrate cores are solid amorphous glass materials.

[0028] In further examples of a package substrate core, the substrate core is a glass core comprising a solid amorphous glass material. The glass substrate core can comprise a glass such as, for example, aluminosilicate, borosilicate, alumino-borosilicate, silica, and fused silica, that additionally optionally comprises one or more of the following: Al_2O_3 , B_2O_3 , MgO , CaO , SrO , BaO , SnO_2 , Na_2O , K_2O , P_2O_5 , ZrO_2 , Li_2O , Ti , and/or Zn . In further examples of glass cores, the glass can comprise silicon and oxygen, as well as optionally any one or more of: aluminum, boron, magnesium, calcium, barium, tin, sodium, potassium, strontium, phosphorus, zirconium, lithium, titanium, and/or zinc. In some examples, a glass package substrate core comprises at least 23% silicon, at least 26% oxygen by weight. In further examples, the glass package substrate core comprises at least 23% silicon, at least 26% oxygen, and at least 5% aluminum by weight.

[0029] Additionally, exemplary solid amorphous glass substrate cores can be considered to have a rectangular prism volume. The rectangular prism volume can contain vias that have been filled with one or more different materials. A material in a via can be a conducting metal such as, for example, copper. Exemplary solid amorphous glass substrate cores can have a thickness in the range of 50 μm to 1.4 mm. Additionally, the package substrate can include a multi-layer glass substrate. The package substrate in this example may be a coreless substrate. The multi-layer glass substrate can have a thickness, for example, in the range of 25 μm to 50 μm . Further, glass substrate cores can have dimensions on a side of 10 mm to 250 mm. For example, the substrate core can be 10 mm by 10 mm up to 250 mm by 250 mm in two dimensions, but substrate cores do not necessarily have to have the same value in both dimensions.

[0030] A package substrate can include one or more interconnect bridges. An interconnect bridge can be partially, fully, or not embedded into the package substrate. An interconnect bridge provides interconnects between chips that are housed on the package substrate. The interconnects can be for I/O between the chips. Some interconnect bridges, such as ones that have conducting through-bridge vias, can also provide power to an operably electrically connected chip. The interconnect bridge can include regions having traces that have a smaller width dimension (the smallest dimension of the trace), a smaller height dimension, and/or a length dimension than the vias and traces of the surrounding package substrate. For example, width dimensions (or smallest dimension) can be 3 μm or less and/or 10 μm or less in some regions. The interconnect bridges can also have smaller trace spacings than the surrounding package substrate. For example, trace center-to-center spacings can be 3 μm and/or less or 10 μm or less in some regions. The bridge can comprise, for example, a silicon substrate, a silicon-on-insulator substrate, one or more organic polymeric materials, a float glass substrate, a borosilicate glass substrate, a silicon dioxide substrate, and/or a silicon nitride substrate. The substrate can additionally comprise, for example, one or more dielectric layers that are comprised of, silicon oxides, silicon nitride, silicon oxynitride, carbon-doped oxide, methyl silsesquioxane, hydrogen silsesquioxane, die backside film (DBF), an epoxy film, a B-stage epoxy film, other dielectric material. The bridge can also be a coreless substrate comprised of a plurality of dielectric layers. The dielectric layers can additionally be, for example, die backside film (DBF), an epoxy film, a B-stage epoxy film, or another dielectric

material. Other materials are possible.

[0031] For packages that include interconnect bridges, the pitch in the interconnect bridge region for first level interconnects (FLIs) assemblies can be less than the pitch for other regions of the FLI assembly. The pitch in the interconnect bridge region for FLIs can be, for example, less than or equal to 25 μm .

[0032] Incorporating through-bridge vias (TBVs) into interconnect bridges can enable power to be routed from a substrate package cavity to a semiconductor device attached to a package substrate. Through-bridge-vias can reduce the number of substrate routing layers required in a package substrate and can result in improved packaging yields. An interconnect bridge having TBVs can be for example, EMIB with TBVs, or EMIB-T. Depending on the bridge substrate material, a TBV may also be described as a through-silicon via (TSV) if the via traverses a region comprised of silicon, for example. However, assembling an interconnect bridge having TBVs into substrate package can present yield challenges and architectures which simplify the assembly process of interconnect bridges having TBVs into a package substrate cavity are important.

[0033] FIGS. 1A-1C illustrate a semiconductor package substrate **100** that includes an interconnect bridge **105**. The interconnect bridge **105** is shown as embedded within package substrate **100**, however the interconnect bridge **105** can also be partially embedded or not embedded in package substrate **100**. The interconnect bridge **105** includes traces and additional vias (that are not depicted for clarity), through-bridge vias **110** (not all are necessarily depicted) and an interconnect bridge substrate **115**. The interconnect bridge substrate **115** can include any of the materials described herein for interconnect bridge substrates, such as for example, silicon-containing substrates, glass substrates, polymer substrates, ceramic substrates. Other types of substrate materials are also possible for the interconnect bridge substrate **115**. Through-bridge vias **110** and traces are comprised of a conducting material, which can be a metal, such as, for example, copper.

[0034] The interconnect bridge **105** is operably electrically connected to the package substrate **100** in an interconnect region through bonding regions **116**. Bonding regions **116** can be, for example, solder regions or can be metal on metal regions formed through thermocompression bonding where solder is not present. Bonding regions **116** connect join-side interconnect bridge pads **117** with interconnect-bridge connection package substrate pads **118**. Join-side interconnect bridge pads **117** can be comprised of a conducting material, such as, for example, copper. Interconnect-bridge connection package substrate pads **118** can be comprised of a metal such as, for example, copper and can optionally comprise one or more surface layers (not shown) comprising a metal or metal alloy, such as, for example, tin and copper, gold, gold and palladium, nickel, and/or platinum. The description interconnect region refers to the connection of the interconnect bridge **110** with the package substrate **100** so that power and/or signals can be delivered to the interconnect bridge **105** from the board-side of the package substrate **100**. (FIG. 1C provides an illustration of a package substrate assembly that includes a board **190**.) The interconnect bridge **110** also includes chip-side interconnect bridge pads **119** which can comprise a conducting material, such as, for example, copper. In this example, the bonding regions **116** are partially surrounded by a portion of dielectric layer region **125** and it can be the case that no underfill material is in the interconnect region between the interconnect bridge **105** and the package substrate **100** (i.e., around bonding regions **116**). An underfill material can be a flowable material such as, for example, an epoxy, a polyimide, or a thermoplastic material. An underfill material is a flowable material, with or without filler particles, that exhibits, for example, good adhesion, thermal stability, and low moisture absorption. The portion of dielectric layer region **125** around bonding regions **116** can provide an insulative barrier between bonding regions **116** that can enhance signal and power delivery integrity by reducing cross talk. Other numbers of and configurations of are also possible for bonding regions **116**, package-side interconnect bridge pads **117**, and embedded-bridge connection package substrate pads **118**.

[0035] In FIGS. 1A-1C, the package substrate **100** also includes a region of collapsible cavities **150**

and **151**. Although some collapsible cavities **151** are illustrated as compressed collapsible cavities **151**, the actual amount of compression of the collapsible cavities **150** and **151** can be more or less than what is illustrated. Collapsible cavities **150** and **151** can be, for example, gas-filled structures that are collapsible. The collapsible cavities **150** and **151** can be in one or more dielectric layers of package substrate **100**. The collapsible cavities **150** and **151** can be, for example, gas-filled glass beads, gas-filled glass fibers, gas bubbles, gas-filled cavities, or SiN or SiOx materials that encase a cavity region filled with a gas. The gas can be, for example, an inert gas, such as for example N.sub.2, or He, or the gas can be air, or a gas comprising N.sub.2 and O.sub.2. Other gases are possible. The collapsible cavities can be fillers in a dielectric layer. The collapsible cavities **150** can also be gas-filled cavities formed through lithographic patterning after laminating a dielectric material. The dielectric layer can be a build-up film or a polyimide dielectric material. The gas-filled glass beads, gas-filled fibers, gas bubbles, gas-filled cavities, or SiN or SiOx materials that encase a cavity region, can have a diameter of, for example, between 3 μm and 30 μm or between 3 μm and 50 μm . FIGS. 2A-2B provide additional examples of alternative collapsible regions that can be in a package substrate. The collapsible regions can be located, for example, in region **160** that is outlined with a dashed line in FIG. 1A.

[0036] The package substrate **100** also includes a package substrate core **120** which can be an organic substrate core or glass substrate core as described herein. Although this package substrate **100** includes a substrate core **120**, coreless substrates are also possible. Additionally, the package substrate **110** has dielectric layer regions **124**, **125** and **126** which can be one or more layers of dielectric (such as build-up layers) having traces and vias **130** and board-side conductive pads **135**. Board-side conductive pads **135** can be connected to a board (e.g., a motherboard, a printed circuit board, a system board, a main board, or a logic board). Chip-side conductive pads **140** and **141** can connect to semiconductor chips and are on an opposite side of the package substrate from the board-side conductive pads **135**. Chip-side conductive pads **140** and **141** can include an optional surface layer **154** and **155**. The surface layer **154** and **155** can be comprised of one or more layers of a metal or metal alloy, comprising, for example, tin, tin and copper, gold, an alloy of gold and palladium, nickel, and/or platinum. Other arrangements and numbers of chip-side conductive pads **140** and **141**, board-side conductive pads **135**, dielectric layer regions **124**, **125** and **126**, and traces and vias **130** are also possible. Traces and vias **130** and board-side conductive pads **135** are comprised of a conducting material which can be a metal, such as, for example, copper.

[0037] FIGS. 1B-1C provide an assembly that includes package substrate **100** and semiconductor devices **170** and **175** that have been electrically connected to package substrate **100**. The semiconductor devices **170** and **175** are electrically connected to package substrate **100** through first level interconnect regions **165** and **166**, that can be, for example, comprised of solder, or can be metal-on-metal interconnections, formed, for example through thermocompression bonding without solder. The first level interconnect regions **165** and **166** join the chip-side conductive pads **140** and **141** of the package substrate **100** with first level interconnect pads (not shown) of the semiconductor devices **170** and **175**. The chip-side conductive pads **140** and **141** can include an optional surface layer **154** and **155**. A dielectric underfill material **180** is between semiconductor devices **170** and **175**, which can be, for example, an epoxy, or thermoplastic material. An underfill material can be a flowable material such as, for example, an epoxy, a polyimide, or a thermoplastic material. An underfill material is a flowable material, with or without filler particles, that exhibits, for example, good adhesion, thermal stability, and low moisture absorption. In FIG. 1C the assembly also includes a board **190** which can be, for example, a motherboard, a printed circuit board, a system board, a main board, or a logic board. The package substrate **100** is electrically connected to the board **190** through solder regions **191**. Dielectric region **192** has cavities that partially surround solder regions **191**. The package substrate **100** can also be electrically connected to board **190** through pins or pads (not shown), for example.

[0038] FIGS. 2A-2B show sections of a package substrate in which the section can be located in

the region **160** outlined by the dashed line in FIG. **1A**. The package substrate sections **200**, **201**, **202**, and **203** include a section of an interconnect bridge **215**. The package substrate sections **204** and **205** include a section of an interconnect bridge **216** and **217**, respectively. The interconnect bridge sections **215**, **216**, and **217** include first join-side interconnect bridge conductive pads **117** and chip-side interconnect bridge conductive pads **119** that are on opposite sides of the interconnect bridges (**215**, **216**, and **217**). The interconnect bridge sections **215**, **216**, and **217** also include interconnect bridge substrate **115**. For elements where the numbering is the same in FIGS. **2A-2B** as in FIGS. **1A-1C**, the description for these elements provided herein for FIGS. **1A-1C** can also be used for these same-numbered elements in FIGS. **2A-2B**. Interconnect bridge substrate **115** can be an interconnect bridge substrate as described herein. Package substrate sections **200**, **201**, **202**, **203**, **204**, and **205** can also be included in assemblies, such as those illustrated in FIGS. **1B** and **1C**. The assemblies can include compressible regions as described by package substrate sections **200**, **201**, **202**, **203**, **204**, and **205** instead of collapsible cavities **150** and **151**, or an assembly can include both. Package substrate sections **200**, **201**, **202**, **203**, **204**, and **205** also include bonding regions **116** and embedded-bridge connection package substrate conductive pads **118**.

[0039] Package substrate sections **200** and **201** include traces **210** that are partially surrounded by gas-filled cavities **220**. In package substrate section **203**, the gas-filled cavities are between traces **210** and embedded-bridge connection package substrate conductive pads **118**. Gas-filled cavities **220** and **221** are compressible regions (or collapsible regions). Gas-filled cavities **220** and **221** are shown as being compressed (partially collapsed) in FIG. **2A**, however, the gas-filled cavities **220** and **221** can be compressed a greater or lesser amount than depicted. Package substrate section **203** of FIG. **2B**, also includes gas-filled cavities **222** that are between embedded-bridge connection package substrate conductive pads **118**. Gas-filled cavities **222** are compressible (or collapsible) regions. The gas-filled cavities **220**, **221**, and **222** can have one or more dimensions of, for example, between 3 μm and 30 μm or between 3 μm and 50 μm . The gas-filled cavities **220**, **221**, and **222** can have other shapes and/or can be more or less deformed than the depictions in FIG. **2A**. Package substrate sections **200-203** can have other numbers and arrangements of traces **210** and gas-filled cavities **220**, **221** and **222**. The gas can be, for example, an inert gas, such as for example N.sub.2, or He, or the gas can be air, or a gas comprising N.sub.2 and O.sub.2. Other gases are possible.

[0040] Package substrate sections **204** and **205** include interconnect bridge sections **216** and **217**, respectively. Interconnect bridge sections **216** and **217** include cavities **230** and **231**, respectively. In package substrate section **204**, one cavity **230** is between two package-join side interconnect bridge conductive pads **117**, and in package substrate section **205**, three cavities **231** are between two package-join side interconnect bridge conductive pads **117**, however other numbers of cavities are also possible, such as two cavities **231** or four cavities **231** between package-join side interconnect bridge conductive pads **117**. Although not depicted, it is also possible that a portion of dielectric layer region **125** can be in cavities **230** and/or **231**. Cavities **230** and **231** can have one or more dimensions that are between 3 μm and 200 μm . Package substrate sections **204** and **205** also include traces **210**, however other numbers and arrangements of traces **210** are also possible.

[0041] In FIGS. **2A-2B**, package substrate sections **200**, **201**, **202**, **203**, **204**, and **205**, the dielectric layer region **125** is partially surrounding bonding regions **116**. Additionally, the package substrate sections **200**, **201**, **202**, **203**, **204**, and **205**, can be assembled so that underfill is not used in the interconnect region between the interconnect bridge **215**, **216**, and **217** and the package substrate **100**. Package substrate sections **200**, **201**, **202**, **203**, **204**, and **205**, can be underfill-less, i.e., there is no underfill material, in the join region between the interconnect bridge **215**, **216**, and **217** and the package substrate **100**. An underfill material can be a flowable material such as, for example, an epoxy, a polyimide, or a thermoplastic material. An underfill material is a flowable material, with or without filler particles, that exhibits, for example, good adhesion, thermal stability, and low moisture absorption. Package substrate sections **200**, **201**, **202**, **203**, **204**, and **205**, can be used

singularly or in various combinations with or without the collapsible cavities **150** described with respect to FIG. **1A** in a semiconductor package substrate **100**.

[0042] FIGS. **3A-3C** illustrate an additional exemplary package substrate **300** and assemblies that include the exemplary package substrate **300**. For elements where the numbering is the same in FIGS. **3A-3C** as it is in FIGS. **1A-1C**, the description provided herein for FIGS. **1A-1C** can also be used for these same-numbered elements in FIGS. **3A-3C**. The package substrate **300** includes an interconnect bridge **305**. The interconnect bridge **305** is shown as embedded within package substrate **300**, however the interconnect bridge **305** can also be partially embedded or not embedded in package substrate **300**. The interconnect bridge **305** includes traces (not depicted for clarity), through-bridge vias **110** (not all are necessarily depicted) and an interconnect bridge substrate **115**. The interconnect bridge **305** also includes join-side interconnect bridge conductive pads **317** that protrude from a surface of the interconnect bridge **305**. Join-side interconnect bridge conductive pads **317** are conducting members that can be, for example, pins or rods having, for example, a rectangular or circular footprint. Other shapes are also possible for the protruding join-side interconnect bridge conductive pads **317**. Join-side interconnect bridge conductive pads **317** can be comprised of a conducting metal, such as, for example, copper. Join-side interconnect bridge conductive pads **317** protrude into cavities in a build-up layer of the build-up layers of dielectric region **325**. The build-up layers can be comprised of, for example, a polyimide or an ABF material. Protruding join-side interconnect bridge conductive pads **317** are electrically connected to the package substrate through bonding regions **116**. Bonding regions **116** can be, for example, solder regions or can be metal on metal regions formed through thermocompression bonding where solder is not present. Embedded-bridge connection package substrate conductive pads **118** can include a layer region **318**. The package substrate pad layer region **318** can be comprised of one or more layers of a metal or metal alloy, comprising, for example, tin, tin and copper, gold, an alloy of gold and palladium, nickel, and/or platinum. It can be the case that there is no underfill present in the interconnect region between interconnect bridge **305** and package substrate **300** (i.e., around Bonding regions **116**). An underfill material can be a flowable material such as, for example, an epoxy, a polyimide, or a thermoplastic material. An underfill material is a flowable material, with or without filler particles, that exhibits, for example, good adhesion, thermal stability, and low moisture absorption.

[0043] FIGS. **3B-3C** provide an assembly that includes package substrate **300** and semiconductor devices **170** and **175** that have been electrically connected to package substrate **300**. The semiconductor devices **170** and **175** are electrically connected to package substrate **100** through first level interconnect regions **165** and **166**, that can be, for example, comprised of solder, or can be metal-on-metal interconnections, formed, for example through thermocompression bonding without solder. In FIG. **3C**, the assembly also includes a board **190** which can be, for example, a motherboard, a printed circuit board, a system board, a main board, or a logic board. The package substrate **100** is electrically connected to the board **190** through solder regions **191**. Dielectric region **192** has cavities that partially surround solder regions **191**.

[0044] FIGS. **4A-4C** provide an additional example of a package substrate **400** that includes an interconnect bridge **305**. Where the numbering of parts is the same for FIGS. **4A-4C** and FIGS. **1A-1C** and **3A-3C**, the descriptions herein with respect to FIGS. **1A-1C** and **3A-3C** are applicable to FIGS. **4A-4C** for the same-numbered parts. The interconnect bridge **305** is shown as embedded within package substrate **400**, however the interconnect bridge **305** can also be partially embedded or not embedded in package substrate **400**. FIG. **4A** illustrates a package substrate **400**, FIG. **4B** illustrates an assembly comprising the package substrate **400** and semiconductor devices **170** and **175**, and FIG. **4C** illustrates an assembly comprising the package substrate **400**, semiconductor devices **170** and **175**, and a board **190**.

[0045] In FIGS. **4A-4C**, the interconnect bridge **305** includes join-side interconnect bridge conductive pads **317** that protrude from a surface of the interconnect bridge **305**. Protruding join-

side interconnect bridge conductive pads **317** are electrically connected to the package substrate through bonding regions **116**. Bonding regions **116** can be, for example, solder regions or can be metal on metal regions formed through thermocompression bonding where solder is not present. Embedded-bridge connection package substrate conductive pads **118** can include a layer region **318**. The package substrate pad layer region **318** can be comprised of one or more layers of a metal or metal alloy, such as, for example, tin, tin and copper, gold, an alloy of gold and palladium, nickel, and/or platinum. Join-side interconnect bridge conductive pads **317** protrude into cavities in a dielectric layer **410** in the dielectric region **425** of packaging substrate **400**. The dielectric layer **410** can be a photo-imageable dielectric, such as for example, polyimide. Dielectric region **425** can be comprised of layers of dielectric material, such as build-up layers. The build-up layers can be comprised of, for example, a polyimide or an ABF material. It can be the case that there is no underfill present between interconnect bridge **305** and package substrate **400**. An underfill material can be a flowable material such as, for example, an epoxy, a polyimide, or a thermoplastic material. An underfill material is a flowable material, with or without filler particles, that exhibits, for example, good adhesion, thermal stability, and low moisture absorption.

[0046] FIG. 5 shows an example configuration for packaged semiconductor devices mounted on a board. Other configurations are possible. In FIG. 5, a board **505** (e.g., a motherboard, a PCB, a system board, a logic board, a circuit board, or a main board) has packaged semiconductor devices **510** and **515** electrically coupled to the board **505**. Interconnect bridges **525** are shown with a dashed line and are covered by packaged semiconductor chips **510** and **515** in this view. The interconnect bridges **525** can be more than one interconnect bridge, and can be interconnect bridges that are with or without TBVs, and semiconductor chip package substrates can contain more than one type of interconnect bridge. The interconnect bridges **525** can be housed in or on package substrates as described by FIGS. 1A, 2A-2B, 3A, and/or 4A and the accompanying description herein. One or more semiconductor devices **510** can be, for example, a processor such as, a central processing unit (CPU), a graphics processing unit (GPU) and/or a field programmable gate array (FPGA), and one or more of the chips **515** can be a high bandwidth memory (HBM) die stack and/or one or more of the chips **515** can transceiver die. HBM can be stacked synchronous dynamic random-access memory SDRAM chips. Other semiconductor devices are also possible.

[0047] Additionally, in FIGS. 1B, 1C, 3B, 3C, 4B, 4C, and 5 the semiconductor devices **170** and **175** can be any combination of microprocessors, CPUs (central processing units), GPUs (graphics processing units), processing cores, system on a chips, other processing hardware, a combination of processors or processing cores, programmable general-purpose or special-purpose microprocessors, accelerators, DSPs, I/O management, programmable controllers, ASICs, programmable logic devices (PLDs), HBM, and/or other memory devices. These semiconductor chip packages can be heterogeneous packages that incorporate different types of chips into one package. The semiconductor chips can be any of the chips, for example, described herein with respect to FIG. 9. The semiconductor chip packages described herein generally can be part of various larger package structures and configurations and the foregoing examples are not meant to limit the types of assemblies that are possible.

[0048] FIGS. 6A-6F provide a method for manufacturing a semiconductor package substrate comprising an interconnect bridge. Where the numbering of parts is the same for FIGS. 6A-6F and FIGS. 1A-1C, the descriptions herein with respect to FIGS. 1A-1C are applicable to FIGS. 6A-6F for the same-numbered parts. In FIG. 6A, a partially manufactured semiconductor substrate **600** comprises a core **120**, dielectric layers **625** and **626**, and traces and vias **631**. Dielectric layers **625** and **626** can be one or more layers of dielectric material, such as ABF. Dielectric layers **625** comprise one or more dielectric layers having collapsible cavities **150** as a filler in the dielectric material. The collapsible cavities **150** can also be gas-filled cavities formed through lithographic patterning after laminating a dielectric material. The collapsible cavities **150** can be, for example, gas-filled glass beads, gas-filled glass fibers, gas bubbles, gas-filled cavities, or SiN or SiOx

materials than encase a cavity region filled with a gas. The gas can be, for example, an inert gas, such as for example N.sub.2, or He, or the gas can be air, or a gas comprising N.sub.2 and O.sub.2. Other gases are possible. Traces and vias **630** comprise a conductive material, such as, for example, copper. Partially manufactured package substrate **601** can be produced by lithographically patterning the frontside and backside surfaces of partially manufactured semiconductor substrate **600** and electrolytically depositing copper to create traces and vias. These processes create traces and vias **630**, interconnect-bridge connection package substrate conductive pads **118**, and sacrificial copper layers **610**. To manufacture partially manufactured semiconductor substrate **602**, a dual lithographic process can be used to pattern and electrolytically deposit tin (layer **615**) and then copper (layer **616**) on the interconnect-bridge connection package substrate conductive pads **118**. [0049] In FIG. **6B**, partially manufactured semiconductor substrate **603** can be produced by a seed etch, a dielectric lamination, and a planarization/reveal process. The dielectric lamination process creates dielectric region **627**. Partially manufactured semiconductor substrate **604** can be created by lithographically patterning and electrolytically depositing an etch stop layer **640**. An etch stop layer **640** comprising copper is optional and can be used for process flows, such as a cavity formation process. Additional buildup layers and vias and traces can be fabricated creating structure **605** in FIG. **6C**, having traces and vias **632** and dielectric region **628**. An excimer laser can be used to create a cavity **645** in partially manufactured semiconductor substrate **605**, creating partially manufactured semiconductor substrate **606** and dielectric region **629**. The cavity creation process can also employ, for example, a CO.sub.2 or a UV laser. Copper layer **640** can act as an etch stop for the excimer laser cavity formation process.

[0050] In FIG. **6D**, partially manufactured semiconductor substrate **607** having dielectric layer **650** can be created through an alkaline etch process in which the tin layer **615** acts as an etch stop. An interconnect bridge **105** is aligned and placed in cavity **645**, as shown by structure **608**. The interconnect bridge **105** is pressed into cavity **645** creating partially manufactured package substrate **609**. The interconnect bridge **105** can be pressed with enough force to cause the buildup layer underneath to deform. The collapsible cavities **150** can absorb stresses. Thermocompression bonding electrically joins the interconnect bridge **105** to the package substrate. Solder can also be used. The remaining semiconductor package **100** features can be manufactured and semiconductor devices **170** and **175** can be attached producing structure **611** of FIG. **6F**. Manufacturing processes include dielectric layer lamination or build-up and formation of traces and vias. FLI pads are also formed. Semiconductor chips can be attached through solder joins or thermocompression bonding joins **165** and **166**. A flip chip process can be used. An underfill material **180** can be flowed into the FLI region between semiconductor devices **170** and **175** and the package substrate **100**.

[0051] FIGS. **7A-7G** illustrate additional methods for manufacturing semiconductor package substrates comprising interconnect bridges. Where the numbering of parts is the same for FIGS. **7A-7G**, FIGS. **3A-3C**, and FIGS. **4A-4C** the descriptions herein with respect to FIGS. **3A-3C** and FIGS. **4A-4C** are applicable to FIGS. **7A-7G** for the same-numbered parts. In FIG. **7A**, a partially manufactured semiconductor substrate **700** comprises a core **120**, dielectric layers **725** and **726**, and traces and vias **730**. Dielectric layers **725** and **726** can be one or more layers of dielectric material, such as a build-up film. Partially manufactured semiconductor substrate **700** also comprises the interconnect-bridge connection package substrate conductive pads **118**. A lithographic process can be used to deposit and pattern a photoresist layer **720** (such as a dry film photoresist) on partially manufactured semiconductor substrate **700**. An electrolytic process can be used to deposit one or more metal or metal alloy surface layers **318** on interconnect-bridge connection package substrate conductive pads **118** creating partially manufactured semiconductor substrate **701**. The one or more surface layers **318** can be comprised of, for example, gold, or a gold/palladium alloy. The photoresist layer **720** can be removed, additional buildup layers can be laminated onto partially manufactured semiconductor substrate **701** and additional conductive traces and vias can be formed to produce partially manufactured semiconductor substrate **702** of FIG. **7B**, having dielectric region

727. An excimer laser skive process can be used to create cavity 745 in partially manufactured semiconductor substrate 703, creating dielectric region 728. A uniform laser beam profile can be used to create a uniform cavity 745 bottom. A desmear process can be used to clean cavity 745 surfaces. An excimer laser process can be used to reveal surface layer 318 on interconnect-bridge connection package substrate conductive pads 118 as shown in partially manufactured semiconductor substrate 703 of FIG. 7C. Dielectric region 729 is created. A dry desmear process can be used to clean the cavity 745 and surface layer 318 surfaces. An interconnect bridge 305 is aligned, placed in cavity 745, and electrically coupled to package substrate 704 in bonding regions 116 which can be solder regions. The remaining semiconductor package 300 features can be manufactured and semiconductor devices 170 and 175 can be attached producing structure 705 of FIG. 7D. Additional manufacturing processes can include filling the cavity 745 between the dielectric region 729 and the interconnect bridge 305 with a dielectric material, such as, for example, a mold material and/or an epoxy material. Additional manufacturing processes can include dielectric layer lamination or build-up and formation of traces and vias. FLI pads are also formed. A flip chip process can be used to attach semiconductor devices 170 and 175. An underfill material 180 can be flowed into the FLI region between semiconductor devices 170 and 175 and the package substrate 300.

[0052] FIGS. 7E-7G provide an alternate process flow starting with partially manufactured semiconductor substrate 702 of FIG. 7B. In FIG. 7E, a laser skive process can be used to create cavity 746 of partially manufactured semiconductor substrate 706. The cavity creation process can employ, for example, a CO.sub.2, an UV (ultraviolet), or an excimer laser. The process creates dielectric region 750. A photo-imageable dielectric can be spray-coated and patterned creating dielectric layer 410 on partially manufactured packaging substrate 707 of FIG. 7F. An interconnect bridge 305 is aligned, placed in cavity 746, and electrically coupled to package substrate 708 in bonding regions 116 which can be solder regions. The remaining semiconductor package 400 features can be manufactured and semiconductor devices 170 and 175 can be attached producing structure 709 of FIG. 7G. Manufacturing processes can include filling the cavity 746 between the dielectric region 750 and the interconnect bridge 305 with a dielectric material, such as, for example, a mold material and/or an epoxy material. Additional manufacturing processes can include dielectric layer lamination or build-up and formation of traces and vias. FLI pads are also formed. Semiconductor chips can be attached through solder joins or thermocompression bonding joins 165 and 166. A flip chip process can be used. An underfill material 180 can be flowed into the FLI region between semiconductor devices 170 and 175 and the package substrate 300.

[0053] FIG. 8 provides a method for manufacturing a semiconductor package substrate that comprises the gas-filled cavities 220, 221, and 222 of FIGS. 2A-2B. A partially manufactured semiconductor substrate is selected 800. The partially manufactured substrate can have a core or be coreless. One or more cavities are formed in one or more layers of dielectric on the partially manufactured substrate 805. Cavities can be formed around one or more traces, for example, to create a package substrate having gas-filled cavities 220, or cavities can be formed in dielectric material to create a package substrate having one or more gas-filled cavities 221 and/or 222. Gas-filled cavities can be formed, for example, through etching, laser drilling, and techniques that employ sacrificial materials. Interconnect-bridge connection package substrate pads can be formed through lithographic patterning and metal deposition processes 810. An interconnect bridge can be placed onto the partially manufactured substrate so that the interconnect bridge pads of the interconnect bridge are aligned with the interconnect-bridge connection package substrate pads 815. The interconnect bridge can be placed into a cavity formed on the package substrate. The cavity can be formed through, for example, processes described with respect to FIGS. 6B-6C. The interconnect bridge can be pressed into the partially manufactured package substrate so that an electrical connection is formed between the interconnect bridge and the partially manufactured substrate 820. FIGS. 6D-6E illustrate, for example, the alignment of an interconnect bridge and

electrical connection formation with the package substrate. Additional manufacturing processes can include forming traces and vias in dielectric layers, forming FLI pads, and attaching semiconductor devices, as described herein and with respect to FIG. 6F, for example.

[0054] FIG. 9 depicts an example computing system. The computing system can be a system used, for example, for running equipment in a semiconductor fabrication plant. For example, instructions for operating processing equipment to perform one or more aspects of the process described in FIGS. 6A-6F and FIGS. 7A-7G can be stored and/or run on the computing system. A computing system 900 can include more, different, or fewer features than the ones described with respect to FIG. 9.

[0055] Computing system 900 includes processor 910, which provides processing, operation management, and execution of instructions for system 900. Processor 910 can include any type of microprocessor, CPU (central processing unit), GPU (graphics processing unit), processing core, or other processing hardware to provide processing for system 900, or a combination of processors or processing cores. Processor 910 controls the overall operation of system 900, and can be or include, one or more programmable general-purpose or special-purpose microprocessors, DSPs, programmable controllers, ASICs, programmable logic devices (PLDs), or the like, or a combination of such devices.

[0056] In one example, system 900 includes interface 912 coupled to processor 910, which can represent a higher speed interface or a high throughput interface for system components needing higher bandwidth connections, such as memory subsystem 920 or graphics interface components 940, and/or accelerators 942. Interface 912 represents an interface circuit, which can be a standalone component or integrated onto a processor die. Where present, graphics interface 940 interfaces to graphics components for providing a visual display to a user of system 900. In one example, the display can include a touchscreen display.

[0057] Accelerators 942 can be a fixed function or programmable offload engine that can be accessed or used by a processor 910. For example, an accelerator among accelerators 942 can provide data compression (DC) capability, cryptography services such as public key encryption (PKE), cipher, hash/authentication capabilities, decryption, or other capabilities or services. In some cases, accelerators 942 can be integrated into a CPU socket (e.g., a connector to a motherboard (or circuit board, printed circuit board, mainboard, system board, or logic board) that includes a CPU and provides an electrical interface with the CPU). For example, accelerators 942 can include a single or multi-core processor, graphics processing unit, logical execution unit single or multi-level cache, functional units usable to independently execute programs or threads, application specific integrated circuits (ASICs), neural network processors (NNPs), programmable control logic, and programmable processing elements such as field programmable gate arrays (FPGAs) or programmable logic devices (PLDs). Accelerators 942 can provide multiple neural networks, CPUs, processor cores, general purpose graphics processing units, or graphics processing units can be made available for use by artificial intelligence (AI) or machine learning (ML) models.

[0058] Memory subsystem 920 represents the main memory of system 900 and provides storage for code to be executed by processor 910, or data values to be used in executing a routine. Memory subsystem 920 can include one or more memory devices 930 such as read-only memory (ROM), flash memory, one or more varieties of random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM) and/or or other memory devices, or a combination of such devices. Memory 930 stores and hosts, among other things, operating system (OS) 932 that provides a software platform for execution of instructions in system 900, and stores and hosts applications 934 and processes 936. In one example, memory subsystem 920 includes memory controller 922, which is a memory controller to generate and issue commands to memory 930. The memory controller 922 can be a physical part of processor 910 or a physical part of interface 912. For example, memory controller 922 can be an integrated memory controller, integrated onto a circuit within processor 910.

[0059] System **900** can also optionally include one or more buses or bus systems between devices, such memory buses, graphics buses, and/or interface buses. Buses or other signal lines can communicatively or electrically couple components together, or both communicatively and electrically couple the components. Buses can include physical communication lines, point-to-point connections, bridges, adapters, controllers, or other circuitry or a combination. Buses can include, for example, one or more of a system bus, a peripheral component interface (PCI) or PCI express (PCIe) bus, a Hyper Transport or industry standard architecture (ISA) bus, a small computer system interface (SCSI) bus, a universal serial bus (USB), or a Firewire bus.

[0060] In one example, system **900** includes interface **914**, which can be coupled to interface **912**. In one example, interface **914** represents an interface circuit, which can include standalone components and integrated circuitry. In one example, user interface components or peripheral components, or both, couple to interface **914**. Network interface **950** provides system **900** the ability to communicate with remote devices (e.g., servers or other computing devices) over one or more networks. Network interface **950** can include an Ethernet adapter, wireless interconnection components, cellular network interconnection components, USB, or other wired or wireless standards-based or proprietary interfaces. Network interface **950** can transmit data to a device that is in the same data center or rack or a remote device, which can include sending data stored in memory.

[0061] Some examples of network interface **950** are part of an infrastructure processing unit (IPU) or data processing unit (DPU), or are used by an IPU or DPU. An xPU can refer at least to an IPU, DPU, GPU, GPGPU (general purpose computing on graphics processing units), or other processing units (e.g., accelerator devices). An IPU or DPU can include a network interface with one or more programmable pipelines or fixed function processors to perform offload of operations that can have been performed by a CPU. The IPU or DPU can include one or more memory devices.

[0062] In one example, system **900** includes one or more input/output (I/O) interface(s) **960**. I/O interface **960** can include one or more interface components through which a user interacts with system **900** (e.g., audio, alphanumeric, tactile/touch, or other interfacing). Peripheral interface **970** can include additional types of hardware interfaces, such as, for example, interfaces to semiconductor fabrication equipment and/or electrostatic charge management devices.

[0063] In one example, system **900** includes storage subsystem **980**. Storage subsystem **980** includes storage device(s) **984**, which can be or include any conventional medium for storing data in a nonvolatile manner, such as one or more magnetic, solid state, and/or optical based disks. Storage **984** can be generically considered to be a “memory,” although memory **930** is typically the executing or operating memory to provide instructions to processor **910**. Whereas storage **984** is nonvolatile, memory **930** can include volatile memory (e.g., the value or state of the data is indeterminate if power is interrupted to system **900**). In one example, storage subsystem **980** includes controller **982** to interface with storage **984**. In one example controller **982** is a physical part of interface **912** or processor **910** or can include circuits or logic in both processor **910** and interface **914**.

[0064] A power source (not depicted) provides power to the components of system **900**. More specifically, power source typically interfaces to one or multiple power supplies in system **900** to provide power to the components of system **900**.

[0065] Exemplary systems may be implemented in various types of computing, smart phones, tablets, personal computers, and networking equipment, such as switches, routers, racks, and blade servers such as those employed in a data center and/or server farm environment.

[0066] An apparatus can comprise a substrate wherein the substrate can comprise layers of dielectric material, wherein the substrate can comprise conductive traces and vias in the layers of dielectric material, and wherein the substrate can comprise one or more dielectric layers that comprise one or more gas-filled cavities; and an interconnect bridge on the substrate wherein the interconnect bridge can comprise conductive traces and conductive through-bridge vias, and

wherein the interconnect bridge is electrically connected to the substrate. The gas-filled cavities can comprise gas-filled glass beads, gas-filled glass fibers, or gas bubbles. In the apparatus, it can be the case that there is no underfill material between the substrate and the interconnect bridge in an interconnect region between the substrate and the interconnect bridge. A gas-filled cavity of the one or more gas-filled cavities can comprise a cavity that partially surrounds a trace in the substrate. The apparatus of claim 1 wherein a gas-filled cavity of the one or more gas-filled cavities has dimensions between 3 μm and 50 μm . The substrate can also comprise a core and the core can be comprised of a solid amorphous glass material.

Examples

[0067] An apparatus can comprise a substrate wherein the substrate can comprise layers of dielectric material, wherein the substrate can comprise conductive traces and vias in the layers of dielectric material, wherein the substrate can comprise first conductive pads, and wherein a layer of the layers of dielectric material can comprise cavities; and an interconnect bridge in the substrate wherein the interconnect bridge comprises conductive traces, conductive vias, and conductive through-bridge vias, wherein the interconnect bridge comprises second conductive pads that protrude from a surface of the interconnect bridge, wherein a second conductive pad protrudes into a cavity of the layer dielectric material that comprises cavities, and wherein a second conductive pad is electrically connected to a first conductive pad. The layer of dielectric material that comprises cavities can be comprised of a polyimide or a build-up film. The layer of dielectric material that comprises cavities can be comprised of a photo-imageable dielectric material. It can be the case that there is no underfill material between the substrate and the interconnect bridge in an interconnect region between the substrate and the interconnect bridge. The first conductive pads can be comprised of copper. The first conductive pads can be comprised of one or more layers of material that comprise tin, gold, palladium, platinum, nickel, or a combination thereof. The second conductive pad can be electrically connected to the first conductive pad through a solder region. The interconnect bridge can be embedded in the substrate. The substrate can comprise a core and the core can be comprised of a solid amorphous glass material.

[0068] An apparatus can comprise a substrate wherein the substrate can comprise layers of dielectric material, wherein the substrate can comprise conductive traces and vias in the layers of dielectric material, wherein the substrate can comprise first conductive pads, and wherein a layer of the layers of dielectric material can comprise cavities; and an interconnect bridge in the substrate wherein the interconnect bridge comprises conductive traces, conductive vias, and conductive through-bridge vias, wherein the interconnect bridge comprises second conductive pads that protrude from a surface of the interconnect bridge, wherein a second conductive pad protrudes into a cavity of the layer dielectric material that comprises cavities, and wherein a second conductive pad is electrically connected to a first conductive pad. The layer of dielectric material that comprises cavities can be comprised of a polyimide or a build-up film. The layer of dielectric material that comprises cavities can be comprised of a photo-imageable dielectric material. It can be the case that there is no underfill material between the substrate and the interconnect bridge in an interconnect region between the substrate and the interconnect bridge. The first conductive pads can be comprised of copper. The first conductive pads can be comprised of one or more layers of material that comprise tin, gold, palladium, platinum, nickel, or a combination thereof. The second conductive pad can be electrically connected to the first conductive pad through a solder region. The interconnect bridge can be embedded in the substrate. The substrate can comprise a core and the core can be comprised of a solid amorphous glass material.

[0069] A method for manufacturing a package substrate can comprise laminating a partially manufactured package substrate with a dielectric material comprising gas-filled cavities; forming interconnect-bridge connection package substrate pads through lithographic patterning and metal deposition processes on the partially manufactured package substrate; placing an interconnect bridge on the partially manufactured package substrate so that interconnect bridge pads of the

interconnect bridge are aligned with the interconnect-bridge connection package substrate pads; and pressing the interconnect bridge into the partially manufactured package substrate so that an electrical connection is formed between the interconnect bridge pads and the interconnect-bridge connection package substrate pads. In the method, the gas-filled cavities can comprise gas-filled glass beads, gas-filled glass fibers, or gas bubbles. It can be the case that, in the method an underfill material is not flowed into an interconnect region between the interconnect bridge and the partially manufactured package substrate. In the method, the interconnect bridge can comprise traces, vias, and through-bridge vias. In the method, the electrical connection can be formed through thermocompression bonding.

[0070] An assembly can comprise a processor and a package substrate, wherein the package substrate comprises: a substrate wherein the substrate comprises layers of dielectric material, wherein conductive traces and vias are in the layers of dielectric material, wherein the substrate comprises first conductive pads, and wherein the substrate comprises a layer of dielectric material that comprises cavities; and an interconnect bridge in the substrate wherein the interconnect bridge comprises conductive traces, conductive vias, and conductive through-bridge vias, wherein the interconnect bridge comprises second conductive pads that protrude from a surface of the interconnect bridge, wherein a second conductive pad protrudes into a cavity of the layer dielectric material that comprises cavities, and wherein a second conductive pads is electrically connected to a first conductive pad, and wherein the processor is electrically coupled to the interconnect bridge. The layer of dielectric material that comprises cavities can be comprised of a build-up film. The layer of dielectric material that comprises cavities can be comprised of a photo-imageable dielectric material. The first conductive pads can comprise one or more layers of material that comprise tin, gold, palladium, platinum, nickel, or a combination thereof. The conductive through-bridge vias are capable of supplying power from a circuit board to the processor.

[0071] Besides what is described herein, various modifications can be made to what is disclosed and implementations of the invention without departing from their scope. Therefore, the illustrations and examples herein should be construed in an illustrative, and not a restrictive sense. The scope of the invention should be measured solely by reference to the claims that follow.

Claims

1. An apparatus comprising: a substrate wherein the substrate comprises layers of dielectric material, wherein the substrate comprises conductive traces and vias in the layers of dielectric material, and wherein the substrate comprises one or more dielectric layers that comprise one or more gas-filled cavities; and an interconnect bridge on the substrate wherein the interconnect bridge comprises conductive traces and conductive through-bridge vias, and wherein the interconnect bridge is electrically connected to the substrate.
2. The apparatus of claim 1 wherein the one or more gas-filled cavities comprise gas-filled glass beads, gas-filled glass fibers, or gas bubbles.
3. The apparatus of claim 1 wherein there is no underfill material between the substrate and the interconnect bridge in an interconnect region between the substrate and the interconnect bridge.
4. The apparatus of claim 1 wherein a gas-filled cavity of the one or more gas-filled cavities comprises a cavity that partially surrounds a trace in the substrate.
5. The apparatus of claim 1 wherein a gas-filled cavity of the one or more gas-filled cavities has dimensions between 3 μm and 50 μm .
6. The apparatus of claim 1 wherein the substrate also comprises a core and the core is comprised of a solid amorphous glass material.
7. An apparatus comprising: a substrate wherein the substrate comprises layers of dielectric material, wherein the substrate comprises conductive traces and vias in the layers of dielectric material, wherein the substrate comprises first conductive pads, and wherein a layer of the layers of

dielectric material comprises cavities; and an interconnect bridge in the substrate wherein the interconnect bridge comprises conductive traces, conductive vias, and conductive through-bridge vias, wherein the interconnect bridge comprises second conductive pads that protrude from a surface of the interconnect bridge, wherein a second conductive pad protrudes into a cavity of the layer dielectric material that comprises cavities, and wherein a second conductive pad is electrically connected to a first conductive pad.

8. The apparatus of claim 7 wherein the layer of dielectric material that comprises cavities is comprised of a polyimide or a build-up film.

9. The apparatus of claim 7 wherein the layer of dielectric material that comprises cavities is comprised of a photo-imageable dielectric material.

10. The apparatus of claim 7 wherein there is no underfill material between the substrate and the interconnect bridge in an interconnect region between the substrate and the interconnect bridge.

11. The apparatus of claim 7 wherein the first conductive pads are comprised of copper.

12. The apparatus of claim 7 wherein the first conductive pads comprise one or more layers of material that are comprised of tin, gold, palladium, platinum, nickel, or a combination thereof.

13. The apparatus of claim 7 wherein the second conductive pad is electrically connected to the first conductive pad through a solder region.

14. The apparatus of claim 7 wherein the interconnect bridge is embedded in the substrate.

15. The apparatus of claim 7 wherein the substrate also comprises a core and the core is comprised of a solid amorphous glass material.

16. An assembly comprising: a processor; and a package substrate, wherein the package substrate comprises: a substrate wherein the substrate comprises layers of dielectric material, wherein conductive traces and vias are in the layers of dielectric material, wherein the substrate comprises first conductive pads, and wherein the substrate comprises a layer of dielectric material that comprises cavities; and an interconnect bridge in the substrate wherein the interconnect bridge comprises conductive traces, conductive vias, and conductive through-bridge vias, wherein the interconnect bridge comprises second conductive pads that protrude from a surface of the interconnect bridge, wherein a second conductive pad protrudes into a cavity of the layer dielectric material that comprises cavities, and wherein a second conductive pads is electrically connected to a first conductive pad, and wherein the processor is electrically coupled to the interconnect bridge.

17. The assembly of claim 16 wherein the layer of dielectric material that comprises cavities is comprised of a build-up film.

18. The assembly of claim 16 wherein the layer of dielectric material that comprises cavities is comprised of a photo-imageable dielectric material.

19. The assembly of claim 16 wherein the first conductive pads comprise one or more layers of material that are comprised of tin, gold, palladium, platinum, nickel, or a combination thereof.

20. The assembly of claim 16 wherein the conductive through-bridge vias are capable of supplying power from a circuit board to the processor.
